

Downloading Our Dataset

Machine learning algorithms find useful patterns in data, so they require a good dataset. We'll use data from the European Social Survey (ESS) for our decision tree. You can download the files we'll use from

<http://bradfordtuckfield.com/ess.csv> and

<http://bradfordtuckfield.com/variables.csv>. (We got our files originally from <https://www.kaggle.com/pascalbliem/european-social-survey-ess-8-ed21-201617>, where they're publicly available for free). The ESS is a large-scale survey of adults across Europe that is conducted every two years. It asks a wide variety of personal questions, including religious affiliation, health status, social life, and level of happiness. The files we'll look at are stored in CSV format. The file extension `.csv` is short for *comma-separated* values, and it's a very common and simple way to store datasets so that they can be opened by Microsoft Excel, LibreOffice Calc, text editors, and some Python modules.

The file `variables.csv` contains a detailed description of each question recorded in the survey. For example, in line 103 of `variables.csv`, we can see a description of a variable called `happy`. This variable records a survey-taker's answer to the question "Taking all things together, how happy would you say you are?" The answers to this question range from 1 (not happy at all) to 10 (extremely happy). Look at the other variables in `variables.csv` to see the variety of information available to us. For example, the variable `sclmeet` records how often respondents meet socially with friends, relatives, or colleagues. The variable `health` records subjective general health. The variable `rlgdgr` records a subjective rating of how religious respondents are, and so on.

After seeing our data, we can start to think of hypotheses related to happiness predictions. We might reasonably suppose that people who have active social lives and good health are happier than others. Other variables—like gender, household size, and age—may be less easy to hypothesize about.

Looking at the Data

Let's start by reading in the data. Download the data from the link and save it locally as `ess.csv`. Then we can use the `pandas` module to work with it,